Correlation Coefficients

Correlation Coefficients

1. Pearson Coefficient

Pearson Correlation is obtained when dividing the covariance value by the standard deviation of both variables.

- Covariance(X, Y) / (σ(X) * σ(Y))

Correlation Coefficients

2. Spearman Coefficient

Covariance(rank(X), rank(Y)) / (σ(rank(X)) * σ(rank(Y)))

Here, rank(X) is the ordered ranking of the feature X in descending order. For example,

X = [10, 15, 9, 12, 5]

rank(X) = [3, 1, 4, 2, 5]

Correlation Coefficients

What is the difference between Pearson and Spearman coefficients?

Pearson coefficient captures the strength of linear dependence between two variables, while Spearman coefficient captures how two variables are related monotonically

Let’s see an example

Correlation Coefficients

Take the function $y = x^3$

x	y	rank(x)	rank(y)
1	1	5	5
2	8	4	4
3	27	3	3
4	64	2	2
5	125	1	1

Pearson’s coefficient =
Covariance(X, Y) / (σ(X) * σ(Y))
~ 0.94

Spearman’s coefficient ~ 1

Correlation Coefficients

Consider two variables x, y whose values vary as shown in the table:

x	1	2	3	4	5	6	7	8	9	10
y	2	4	6	8	10	12	14	16	18	20

Without using any math, you can see that y is just double x. So this means they are highly correlated.

Correlation Coefficients

Mean of x = $\mu_x = 5.5$
Mean of y = $\mu_y = 11$

x	y	xi - µx(=5.5)	yi - µy(=11)	(xi - µx)(yi - µy)
1	2	-4.5	-9	40.5
2	4	-3.5	-7	24.5
3	6	-2.5	-5	12.5
4	8	-1.5	-3	4.5
5	10	-0.5	-1	0.5
6	12	0.5	1	0.5
7	14	1.5	3	4.5
8	16	2.5	5	12.5
9	18	3.5	7	24.5
10	20	4.5	9	40.5

Correlation Coefficients

$\sum (x_i - \mu_x)(y_i - \mu_y) = 165$

Covariance = $\sum (x_i - \mu_x)(y_i - \mu_y) / (n - 1) = 165/9 = 18.33$

To calculate the correlation coefficient, all we need is the standard deviation of x and y.
Standard deviation of x = 3.028
Standard deviation of y = 6.055

Correlation coefficient = 18.33/(3.028*6.055) = 1.

Correlation Coefficients

Using the previous example, what do you think will be the covariance and correlation between x and y if we change values of y to 3 times of x rather than two times of x?

x	1	2	3	4	5	6	7	8	9	10
y	3	6	9	12	15	18	21	24	27	30

(A) Covariance and correlation both will remain the same
(B) Covariance increases, the correlation remains the same
(C) Correlation and correlation both will increase
(D) Covariance increases, but the correlation will decrease.

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